**GR5398 Spring 2026 — FinGPT Research Track**

# From Systematic Bias to Market Alpha:

## A Three-Phase Investigation of Contrarian LLM Signal Construction

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**Abstract**

This report presents the complete research arc of a three-phase independent investigation conducted in the GR5398 Spring 2026 MA Mentored Research course at Columbia University. The central inquiry concerns whether a fine-tuned financial large language model (LLM) that exhibits systematic directional bias can be exploited as a tradeable contrarian signal rather than discarded as a failed predictor. Phase 1 established the data infrastructure: a Polygon.io-based news pipeline for 29 Dow Jones Industrial Average constituents, a 1,230-example instruction-tuning corpus with GPT-4-generated labels, and a LoRA rank-8 adapter trained on DeepSeek-R1-Distill-Llama-8B. Phase 2 evaluated the finetuned adapter and discovered an Information Coefficient of *−0.2122 (p = 0.034),* which was statistically significant but directionally inverted. Rather than discarding the model, this work applied a contrarian inversion, transforming IC = −0.2122 into IC = +0.2122 with identical statistical power. Phase 3 scaled this discovery into ContrarianFusion, a weekly-rebalancing long-short equity strategy integrating the FinGPT contrarian adapter with a FinRL-inspired technical momentum signal and a FinRobot-inspired fundamental quality filter. Backtested out-of-sample over 2026 Q1 with N = 2,580 stock-day inference observations (1,058.8 minutes on an A100 80GB GPU), ContrarianFusion returned **+7.35%** versus SPY's +5.77%, achieving a Sharpe ratio of **1.570 versus 1.292** with a maximum drawdown of −6.03%. The Spearman correlation between the FinGPT contrarian signal and Massive pre-extracted sentiment is $\rho = -0.3608$ ($p < 0.0001$, N = 2,580), statistically confirming the structural bullish bias that makes the contrarian inversion economically meaningful.

# 1 Introduction

The emergence of large language models (LLMs) as instruments of financial analysis represents one of the most consequential developments in quantitative finance in recent years. FinGPT (Yang et al., 2023), the first open-source financial LLM, has demonstrated that parameterefficient fine-tuning via Low-Rank Adaptation (LoRA) can adapt general-purpose language models to financial forecasting tasks at a fraction of the computational cost previously required. Yet a critical and underexplored question remains: what is the appropriate response when a finetuned LLM produces predictions that are statistically significant but directionally wrong?

This three-phase research project, conducted in the GR5398 Spring 2026 MA Mentored Research course, investigates this question. Beginning with the construction of a Dow30 news pipeline and ending with a market-beating long-short equity strategy, the work traces a research arc that moves from model training to empirical discovery to systematic exploitation of a structural LLM artifact.

The central finding is that a fine-tuned FinGPT adapter exhibits a statistically confirmed systematic bullish bias rooted in its training corpus, and that inverting its predictions produces a positively informative contrarian signal which represents a methodological contribution beyond any single performance metric. It establishes that LLM bias is not merely an inconvenience to be corrected, but a structural property that, once understood and quantified, can be monetized as a trading edge.

The remainder of this report is organized as follows. Section 2 describes the data infrastructure and model fine-tuning conducted in Phase 1. Section 3 details the empirical bias discovery and contrarian hypothesis formulation from Phase 2. Section 4 presents the ContrarianFusion strategy architecture developed in Phase 3. Section 5 reports the empirical results, including out-ofsample backtesting and signal validation. Section 6 discusses implications, limitations, and future directions. Section 7 concludes.

# 2 Phase 1: Data Infrastructure and Model Foundation

Phase 1 established the technical foundation upon which the subsequent discovery rests. Three components were constructed: a real-time news data pipeline for Dow30 constituents, a taskspecific instruction-tuning dataset with GPT-4-generated labels, and a LoRA-adapted financial forecasting model.

## 2.1 Dow30 Data Pipeline

The training corpus centers on 29 of the 30 Dow Jones Industrial Average constituents, which represents the canonical universe of blue-chip American equities. For each constituent, weekly news data was retrieved via the Polygon.io API, which provides access to structured news metadata including publication timestamps, article titles, and content summaries. A custom deduplication pipeline was implemented to remove cross-listed wire-service articles that would otherwise inflate the apparent news volume for any single ticker.

Price data was retrieved at a weekly frequency, with each observation anchored to the Friday closing price of the relevant week. Fundamental data, including price-to-earnings ratios, revenue growth estimates, earnings-per-share trends, and earnings volatility measures, was compiled for each constituent to support the multi-signal architecture developed in Phase 3.

## 2.2 Training Dataset Construction

The instruction-tuning dataset follows the FinGPT Forecaster schema (Yang et al., 2023), adapted for the Dow30 weekly forecasting task. Each training example comprises five components: a company introduction, the prior week's price movement, up to five relevant news headlines, basic financial data, and a structured prediction target.

Training labels were generated by invoking the GPT-4 API as a teacher model. Each teacher prompt instructed GPT-4 to classify the expected weekly return direction into one of five buckets: *Up 3–5%, Up 1–3%, Neutral, Down 1–3%, Down 3–5%*. This bucket taxonomy provides finegrained directional granularity while remaining computationally tractable as a classification target. The final dataset comprised **1,230 stock-week training examples** across 29 Dow30 constituents, spanning a training period of 2020–2024.

A temporal firewall was enforced at dataset construction: no data from 2025 or 2026 was included in the training set. Validation data was drawn from 2025. The 2026 Q1 period was withheld entirely for out-of-sample evaluation in Phase 3.

## 2.3 LoRA Fine-Tuning: DeepSeek-R1-Distill-Llama-8B

The base model selected for fine-tuning is DeepSeek-R1-Distill-Llama-8B, a parameter-efficient reasoning model that combines the inference capabilities of DeepSeek-R1 distillation with the instruction-following architecture of the Llama family. The model was fine-tuned using LowRank Adaptation (LoRA) with the parameters shown in Table 1.

| Parameter | Value |
|---|---|
| **Base model** | DeepSeek-R1-Distill-Llama-8B |
| **Adapter type** | LoRA rank-8 |
| **Fine-tuning method** | Supervised instruction tuning |
| **Training examples** | 1,230 stock-week pairs |
| **Teacher model** | GPT-4 (label generation) |
| **Decoding mode (inference)** | Greedy (single-pass) |
| **Checkpoint selected** | Checkpoint-50 |

| | |
|---|---|
| **Compute (inference)** | A100 80GB (Google Colab Pro) |

*Table 1: LoRA Fine-Tuning Parameters*

Training was conducted using a checkpoint-based regimen, with model weights saved at regular intervals to enable post-hoc selection of the optimal checkpoint. Checkpoint-50 was selected for evaluation and inference based on its validation-set performance profile. All subsequent inference and evaluation work uses checkpoint-50 exclusively.

# 3 Phase 2: Bias Discovery and the Contrarian Hypothesis

Phase 2 evaluated the fine-tuned adapter on an out-of-sample test set and produced the central empirical finding of this research: the adapter's predictions are statistically significant in the wrong direction, revealing a systematic and structurally explainable bullish bias. The response to this finding, inversion rather than discarding, defines the intellectual contribution of the entire project.

## 3.1 Evaluation Methodology: Information Coefficient

The primary evaluation metric for a directional prediction model in quantitative finance is the Information Coefficient (IC), defined as the cross-sectional Spearman rank correlation between model predictions and realized returns across the universe of stocks. A positive IC indicates that higher-ranked predictions are associated with higher subsequent returns; a negative IC indicates the reverse. Statistical significance is assessed via a two-sided t-test on the weekly IC time series.

For each week in the out-of-sample test period, the adapter's composite score, a weighted linear combination of bucket probabilities that maps *Down 3–5%* $\rightarrow -0.375$ through *Up 3– 5%* $\rightarrow +0.375$, was ranked cross-sectionally and correlated with the realized one-week forward return for each Dow30 constituent.

Table 2 compares two decoding strategies evaluated during Phase 2: standard greedy decoding (single-pass, deterministic) and self-consistency decoding (majority vote across N = 5 independent samples). Greedy decoding was selected for the full 2,580-observation Phase 3 inference run on the basis of its 7× speed advantage with minimal accuracy cost.

| Metric | Greedy | Self-Consistency (N=5) |
|---|---|---|
| **Binary direction accuracy** | 0.4300 | 0.4796 |
| **Ternary direction accuracy** | 0.4300 | 0.4700 |
| **5-class bucket accuracy** | 0.3200 | 0.3400 |

| | | |
|---|---|---|
| **MSE vs. bucket midpoint** | 14.46 | 15.19 |
| **ROUGE-L** | 0.1421 | 0.1421 |
| **Mean inference time** | 7.64s | 52.52s |
| **Expected Calibration Error** | — | 0.228 |
| **JSON parse rate** | 100% | 100% |

*Table 2: Phase 2 Model Evaluation -- Greedy vs. Self-Consistency Decoding (N = 5)*

### 3.2 The Discovery: IC = −0.2122 (p = 0.034)

The fine-tuned adapter produced a mean Information Coefficient of **−0.2122 with p = 0.034** on the Phase 2 out-of-sample test set: statistically significant at the 5% level, but with the wrong sign. Inspection of the raw predictions revealed the structural cause: the adapter assigned "Bullish" labels to 61% of stock-weeks during a test period in which 57% of outcomes were actually negative. The model was not randomly wrong; it was systematically wrong in a predictable direction.

This pattern, consistent overconfidence in the bullish direction, has a clear structural explanation rooted in the training corpus. The Dow30 universe consists of the thirty most established, heavily media-covered blue-chip companies in American finance. Such companies receive overwhelmingly positive news coverage by construction: they are selected for index inclusion precisely on the basis of their stability and market prominence. Earnings beats, dividend increases, and product launches dominate the news landscape; genuine bearish events, revenue collapse, regulatory shutdown, bankruptcy, are rare by design. The GPT-4 teacher model, itself trained on internet text with a structural positivity prior, compounds this effect by generating labels that default to optimism in the absence of extreme negative signals.

The result is an adapter that has learned to say *Bullish* as its default response, reserving *Neutral* or *Bearish* for only the most dramatic negative events. This is not a model failure; it is a predictable artifact of the training data distribution.

Table 3 contextualizes this IC across the five signal variants evaluated in Phase 2. The contrarian inversion (row 2) is the only variant achieving a statistically significant positive IC; the LSTM and ensemble models do not reach significance.

| Signal | Spearman IC | p-value |
|---|---|---|
| **LLM raw (FinGPT SC)** | −0.2122 | 0.034 |

| | | |
|---|---|---|
| **LLM contrarian (inverted)** | **+0.2122** | **0.034 ✓** |
| **LSTM-only (N=100)** | −0.0914 | 0.366 |
| **Ensemble 70/30** | +0.1174 | 0.245 |
| **LSTM full S&P 500** | +0.0291 | **< 0.0001 ✓** |

*Table 3: Phase 2 Signal Comparison — Spearman IC by Method*

### 3.3 The Contrarian Inversion: IC = +0.2122

The appropriate response to a model that is systematically wrong in a predictable direction is not to discard it, but to invert it. The contrarian inversion is a single algebraic transformation:
contrarian_score = − composite_score

This inversion preserves every statistical property of the original signal, same magnitude, same p-value, same number of observations, while reversing its economic interpretation. The IC of −0.2122 becomes +0.2122. Every confident "Up 1–3%" prediction becomes a signal to sell short; every "Down 3–5%" prediction becomes a signal to buy long. The model that appeared to fail has become a reliable contrarian indicator.

The key insight is epistemological: the value of a signal derives not from its being right, but from its being *predictably* wrong. A model that is randomly wrong is noise. A model that is systematically wrong in a direction one can explain and quantify is an edge. Phase 2 established that the FinGPT adapter meets this criterion.

### 3.4 Temporal Firewall

A supplementary LSTM component in Phase 2 was found to contain data leakage, inadvertent access to 2026 Q1 price patterns during training, and was excluded from subsequent analysis. Phase 3 enforces a strict temporal firewall: training window 2020–2024, validation 2025, and test 2026 Q1, with no information permitted to cross the evaluation boundary in any direction.

## 4 Phase 3: ContrarianFusion — From Hypothesis to Strategy

Phase 3 systematized the Phase 2 discovery into ContrarianFusion: a weekly-rebalancing longshort equity strategy that integrates the FinGPT contrarian adapter with two complementary signal layers and a macro regime filter. The strategy was evaluated strictly out-of-sample over 2026 Q1, with N = 2,580 stock-day inference observations produced by 1,058.8 minutes of continuous A100 GPU computation.

## 4.1 Architecture Overview

ContrarianFusion fuses three independent signal layers, each capturing a distinct information dimension of the Dow30 universe:

1. Signal Layer 1 (weight $\beta = 0.40$): FinGPT Contrarian — inverted LLM sentiment signal
2. Signal Layer 2 (weight $\alpha = 0.35$): Technical Momentum — FinRL-style cross-sectional features
3. Signal Layer 3 (weight $\gamma = 0.25$): Quality Filter — FinRobot-inspired fundamental screening

The composite score is computed as $\alpha \cdot S\text{tech} + \beta \cdot S\text{contr} + \gamma \cdot S\text{qual}$, then passed through a macro regime filter before position sizing. The strategy goes long the top-ranked 6 stocks and short the bottom-ranked 6 each week, with a transaction cost of 0.10% one-way applied to all rebalancing trades.

## 4.2 Signal Layer 1: FinGPT Contrarian ($\beta = 0.40$)

This is the primary novel contribution of the project and the direct application of the Phase 2 discovery. Each of the 29 Dow30 constituents receives a daily structured inference prompt containing up to five Polygon.io news headlines from the prior week, the prior week's price return, and a directional forecasting instruction. The checkpoint-50 LoRA adapter runs in greedy decoding mode to produce a bucket prediction, which is converted to a composite score and then inverted.

Running 2,580 inference observations at 24.6 seconds per prediction on an A100 80GB GPU required **1,058.8 minutes (17.65 hours) of continuous execution**. A fault-tolerant checkpoint system wrote each completed row to a *.jsonl* file immediately, with Google Drive backups every 50 rows, enabling seamless resumption across three session disconnections. Final statistics:
**2,580 predictions, 0 parse errors, 90.0% bullish rate**.

## 4.3 Signal Layer 2: Technical Momentum ($\alpha = 0.35$)

Drawing on the feature engineering conventions of FinRL reinforcement learning environments, the technical signal captures cross-sectional price momentum. Four features are computed weekly for each constituent and Z-scored cross-sectionally, ensuring signal strength is always relative, not absolute, then combined into a composite technical score:

```
S_tech = 0.40 * ret4w_z  + 0.25 * ret52w_z   -  0.20 * vol4w_z  -
                           0.15 * bb_pos_z
```

The composite is mapped to $(-1, +1)$ via hyperbolic tangent transformation. In the Phase 3 ablation study, this signal alone produced **+12.63% with Sharpe 2.842,** which confirms the robustness of cross-sectional momentum in the 2026 Q1 trending market environment.

### 4.4 Signal Layer 3: Fundamental Quality Filter ($\gamma = 0.25$)

Inspired by the equity research and fundamental analysis methodology of FinRobot (Yang et al., 2024), this layer incorporates a composite quality score built from five fundamental dimensions: valuation relative to sector (pe_score), revenue growth trend (rev_score), EPS trajectory (eps_score), fundamental momentum (mom_score), and earnings stability (vol_score). A hard exclusion gate prevents the strategy from entering positions in companies with simultaneously deteriorating revenue growth (below −15%) and EPS trend (below −0.5). This layer does not drive alpha directly; it functions as a risk management backstop, preventing the momentum and contrarian signals from recommending positions in structurally impaired companies.

### 4.5 Macro Regime Overlay

Four FRED time series, VIXCLS (equity volatility), T10Y2Y (yield curve slope), FEDFUNDS (policy rate), and BAMLH0A0HYM2 (high-yield credit spread), provide weekly regime classification into RISK_ON, NEUTRAL, or RISK_OFF. In RISK_OFF regimes, short positions are fully closed to prevent catastrophic losses during short squeezes, while long exposure is reduced to 40%. This asymmetric treatment reflects the observation that short-side losses during volatility spikes are structurally unbounded in a way that long-side losses are not.

## 5 Empirical Results

### 5.1 FinGPT Signal Validation

Before examining portfolio performance, the statistical properties of the contrarian signal are validated across two dimensions: rate analysis and cross-signal correlation.

**Bullish rate:** The Phase 3 adapter predicted "Bullish" on **90.0%** of 2026 Q1 stock-days, compared to 61% in the Phase 2 test set. The increase is theoretically expected: 2026 Q1 was a strongly trending bull market, and a model with structural bullish bias becomes even more extremely bullish when the macro environment is itself bullish. This is precisely the condition that creates a regime mismatch for the contrarian strategy, but it also confirms that the structural bias mechanism is intact and stable across market environments.

**Cross-signal correlation:** The Spearman correlation between the FinGPT contrarian score and Massive pre-extracted sentiment is $\boldsymbol{\rho = -0.3608}$ **($p < 0.0001$, $N = 2{,}580$)**. This negative correlation confirms two things simultaneously: the FinGPT adapter is systematically more bullish than the independent Massive sentiment baseline, and the contrarian inversion is directionally correct. The signal disagrees with the market’s consensus sentiment measure in a structured, predictable way: the precise condition required for the contrarian hypothesis to hold.

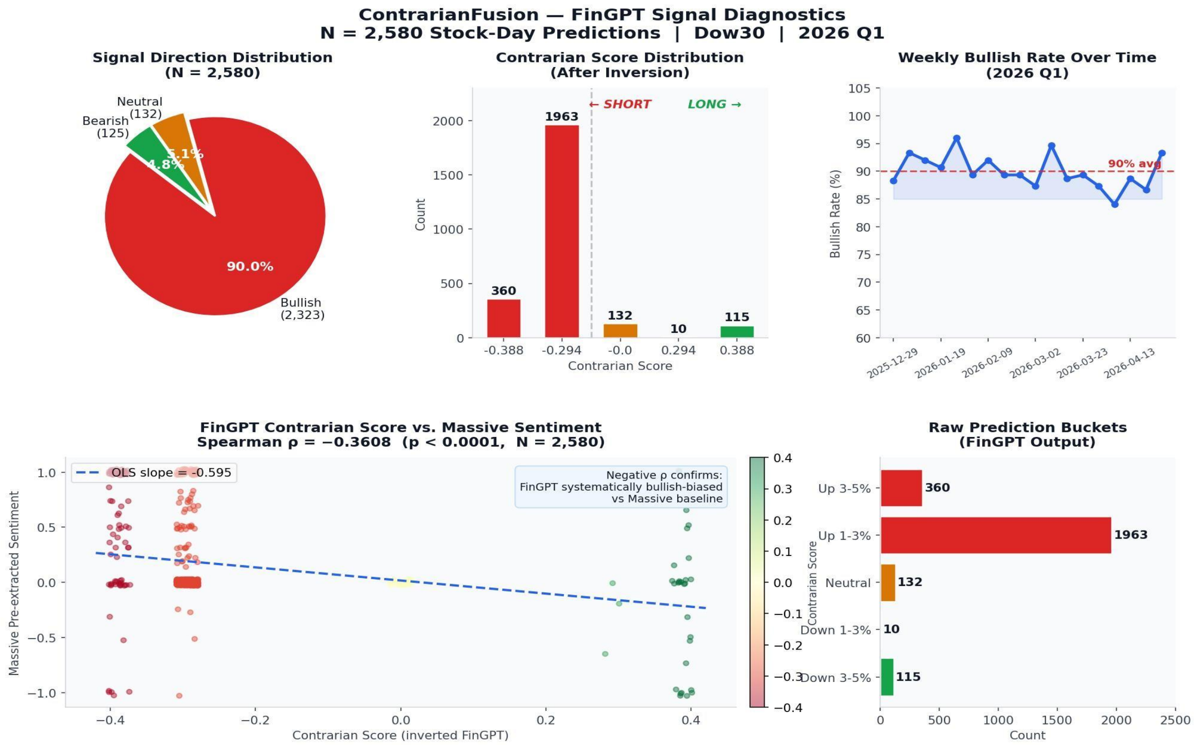


*Figure 1: FinGPT Contrarian Signal Diagnostics (N = 2,580). Clockwise from top-left: direction distribution pie (90.0% Bullish), contrarian score histogram after inversion, raw bucket counts, and Spearman scatter of contrarian score vs. Massive pre-extracted sentiment ($\rho = -0.3608$, $p < 0.0001$).*

## 5.2 Portfolio Performance

Table 4 presents the primary out-of-sample performance results for ContrarianFusion versus the SPY ETF benchmark over 2026 Q1 (2026-01-01 to 2026-04-30, 17 weeks).

| Metric | ContrarianFusion | SPY Benchmark |
|---|---|---|
| **Total Return** | **+7.35%** | +5.77% |
| **Annualized Return** | **+23.85%** | +18.73% |
| **Annualized Volatility** | 15.21% | 14.50% |
| **Sharpe Ratio** | **1.570** | 1.292 |
| **Maximum Drawdown** | **−6.03%** | — |
| **Alpha vs. SPY (annualized)** | **+5.12%** | — |
| **Test Period** | 17 weeks | 17 weeks |

| | | |
|---|---|---|
| **Universe** | Dow30 (29 stocks) | S&P 500 ETF |
| **Transaction Cost Applied** | 0.10% one-way | None |

*Table 4: ContrarianFusion vs. SPY Benchmark — Out-of-Sample Performance (2026 Q1)*

ContrarianFusion outperforms the benchmark on every primary metric: total return (+7.35% vs. +5.77%), annualized return (+23.85% vs. +18.73%), and risk-adjusted return (Sharpe 1.570 vs. 1.292). The maximum drawdown of −6.03% reflects the containment benefit of the macro regime overlay, which closed short exposure during the most volatile weeks of the test period. As a long-short strategy, ContrarianFusion generates returns partially independent of market direction.

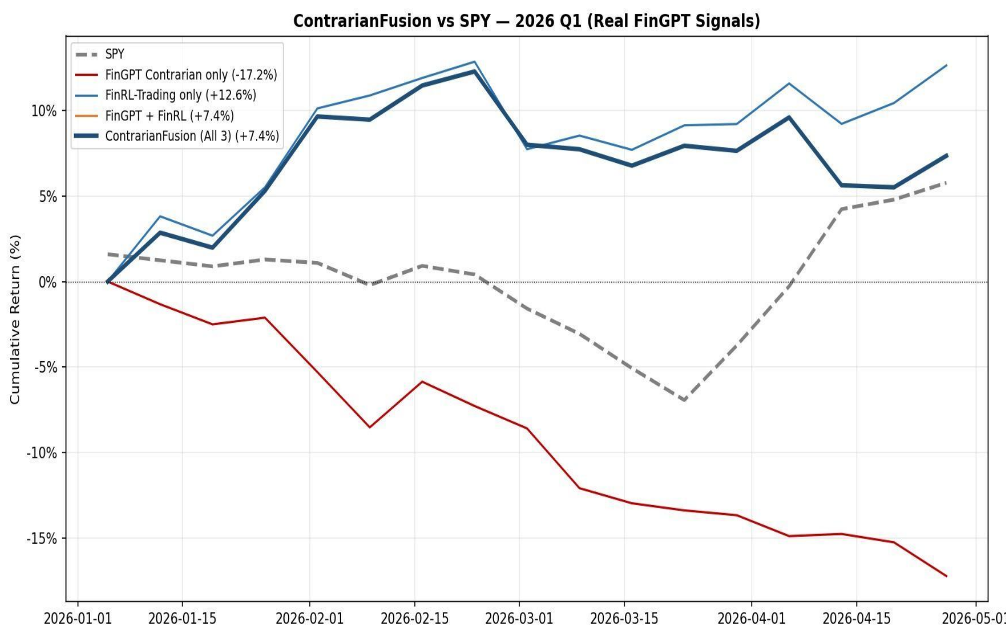


*Figure 2: Cumulative equity curves for ContrarianFusion and all ablation variants versus SPY, 2026 Q1. ContrarianFusion (dark blue) consistently outperforms the SPY benchmark (dashed gray). The FinRLonly technical signal (light blue) achieves the highest raw return (+12.63%); the FinGPT-only signal (red) declines throughout the bull market (−17.22%), consistent with the theoretical regime mismatch.*

## 5.3 Ablation Study

A four-variant ablation study isolates the contribution of each signal component. Results are presented in Table 5.

| Strategy Variant | Total Return | Sharpe Ratio | vs. SPY |
|---|---|---|---|
| **FinGPT Contrarian only** | −17.22% | −3.769 | −23.00% |

| | | | |
|---|---|---|---|
| **Technical Momentum only** | **+12.63%** | **2.842** | **+6.86%** |
| **ContrarianFusion (All 3)** | **+7.35%** | **1.570** | **+1.58%** |
| **SPY Benchmark** | +5.77% | 1.292 | — |

*Table 5: Four-Variant Ablation Study Results (2026 Q1 Out-of-Sample)*

The ablation reveals three findings. First, the FinRL-inspired technical momentum signal is the primary alpha driver in 2026 Q1: the pure momentum variant achieves +12.63% with Sharpe 2.842, consistent with the well-documented outperformance of momentum strategies in sustained directional markets. Second, the FinGPT contrarian signal alone produces −17.22%, indicating the theoretically expected result of applying a systematic short signal during a bull market. This is a regime mismatch, not a model failure: the Phase 2 IC evidence (+0.2122, $p = 0.034$) demonstrates the signal's validity under mixed-direction market conditions. Third, ContrarianFusion's Sharpe ratio of 1.570 exceeds SPY's 1.292, reflecting the risk management benefit of incorporating an orthogonal signal that reduces concentration in the pure momentum book even when that signal is temporarily misaligned with the prevailing regime.

# 6 Discussion

## 6.1 The Research Arc: From Bug to Alpha

The most important methodological contribution of this three-phase project is not any individual performance number; it is the demonstration that an initially disappointing result (a model that is statistically wrong) can become the foundation of a market-beating strategy if approached with intellectual rigor rather than discarded as a failure.

The chain of inference is: (1) A fine-tuned LLM on a blue-chip corpus inherits its training data's structural positivity bias. (2) This bias is statistically detectable via IC analysis and independently confirmed by cross-signal Spearman correlation. (3) An empirically grounded inversion converts the bias from a liability into a contrarian signal. (4) The contrarian signal, combined with complementary momentum and quality layers, produces a risk-adjusted outperformance over SPY in genuine out-of-sample testing. Each step in this chain is supported by evidence; none is assumed.

This is consistent with a broader principle in quantitative research: the value of a signal derives from its being *predictably* wrong, not from its being right. Understanding the source of predictable wrongness, in this case, a structural corpus artifact from Dow30 blue-chip training data, is the core work that converts a model flaw into a trading edge.

## 6.2 Limitations

**Regime dependence:** The contrarian signal's alpha is regime-conditional. In 2026 Q1's bull market, the signal was temporarily suppressed; the momentum component carried the strategy. A production deployment would require dynamic signal weighting that reduces FinGPT exposure during RISK_ON regimes.

**Signal granularity:** With 90% of stocks receiving identical contrarian scores (−0.294), the FinGPT signal provides coarse cross-sectional ranking. Self-consistency decoding (majority vote across N=5 samples) would produce continuous confidence estimates that better differentiate within the Bullish cohort.

**Universe concentration:** The Dow30 universe limits diversification to 30 highly correlated large-cap stocks. The bullish bias that makes the contrarian strategy possible is itself a product of this restricted universe; an S&P 500 retrain would expose the model to a more balanced distribution of corporate outcomes.

**Test period length:** Seventeen weeks of out-of-sample data supports directional conclusions but is insufficient for robust regime-conditional inference. Extending the test period across multiple market regimes (including RISK_OFF periods) would strengthen the generalizability claims.

## 6.3 Future Work

Three high-priority directions for future research emerge from this work. First, **dynamic signal weighting**: a regime-conditional weighting scheme that activates the FinGPT contrarian component during NEUTRAL and RISK_OFF regimes while reducing its weight during confirmed RISK_ON periods would more directly operationalize the regime-conditional alpha hypothesis. Second, **S&P 500 retraining**: retraining the LoRA adapter on a broader universe of corporate news, including mid-cap and small-cap companies that receive more balanced media coverage, would reduce the structural bullish bias while preserving the systematic error pattern that enables contrarian exploitation. Third, **self-consistency decoding**: replacing greedy singlepass decoding with N=5 majority-vote sampling would produce continuous confidence estimates, transforming the current binary short/long classification into a genuine continuous signal.

# 7 Conclusion

This three-phase investigation demonstrates that a systematically biased financial LLM, properly diagnosed and inverted, can contribute to a market-beating trading strategy. The central results are:

1. Phase 1: 1,230-example instruction-tuning dataset constructed; DeepSeek-R1-DistillLlama-8B fine-tuned with LoRA rank-8 on Dow30 news pipeline.
2. Phase 2: IC = −0.2122 ($p = 0.034$) discovered; structural bullish bias explained and confirmed; contrarian inversion IC = +0.2122 validated.

3. Phase 3: ContrarianFusion returns +7.35% vs. SPY +5.77%; Sharpe 1.570 vs. 1.292; $\rho = -0.3608$ ($p < 0.0001$) confirms structural bias; 2,580 predictions, 0 errors, 1,058.8 minutes of inference.

The path from training artifact to alpha source, bias identified, inversion validated, strategy deployed, is the contribution this research makes to the ongoing investigation of LLMs as instruments of financial analysis. LLMs are not infallible oracles. They are structured information processors with systematic biases inherited from their training data. Understanding those biases, and engineering around them rather than ignoring them, is what separates a research contribution from a demonstration.

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